Comprehensive PDF Extraction Benchmark

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1 Standard Text Layer

This section contains standard, selectable text. Your pipeline should easily extract this paragraph. The goal is to establish a baseline for text-to-coordinate mapping. Recent studies [1, 2] suggest that hybrid parsers outperform pure OCR models in structured documents.

The first mathematical test is the Quadratic Formula, which includes a square root and a fraction:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{1}$$

2 Tabular and Visual Data

Below is a table with mixed data types. Testing the structural integrity of your output (CSV/JSON/Markdown) is key here. As referenced in the data standards manual [3], cell alignment must be preserved.

Test ID	Status	Confidence Score
TX-900	PASS	0.998
TX-901	FAIL	0.450
TX-902	PENDING	N/A

Table 1: Validation results for automated parsing units.

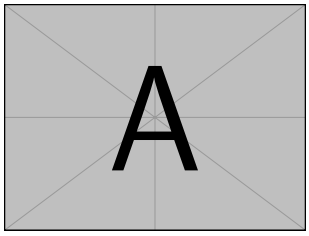

Figure 1: Reference image for spatial bounding box detection.

The second formula tests Greek letters and summation limits:

$$\Phi = \sum_{i=1}^{n} \frac{\alpha_i}{\beta_i + \gamma}$$

3 Conclusion and References

Our final mathematical test is the definition of a limit, placed just before the bibliography to test proximity parsing:

$$\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x = e$$

Success is defined by the accurate extraction of the multi-entry bibliography below and the correct resolution of internal links.

References

[1] Alpha, B. & Gamma, D. (2026). *Heuristics for Non-Standard PDF Parsing.* OCR Quarterly, 22(1).

[2] Smith, J., et al. (2025). *The Evolution of Neural Document Analysis.* AI Review, 15(4), 200-215.

[3] ISO Standard 32000-2. (2024). *Document Management — Portable Document Format — Part 2: PDF 2.0.*

[4] Zhang, L. (2023). *Challenges in Table Extraction from Unstructured Data.* Data Engineering Today.

4 The Glyph Challenge (OCR Test)

The following image is a screenshot of a text block. It contains zero digital text layer information. If your pipeline relies solely on the PDF text stream, this section will return empty.

3 The Glyph Challenge (OCR Test)

The following paragraph is rendered as a vector graphic. It lacks a standard ‘ToUnicode‘ mapping. If you try to select this text in a standard PDF viewer, it may behave like an image. Your pipeline must use OCR to ”read” this:

> This paragraph is a ”pure glyph” test. It is contained within a TikZ node, effectively treating the text as a series of paths or a graphical object rather than a standard text stream. If your pipeline relies solely on the PDF text layer, this section will return as an empty string or null. Successful extraction proves your system can trigger OCR when the text layer is missing or obfuscated.

1

Figure 2: Rasterized text for visual extraction testing.